SIH26262 Master Execution & Architecture Plan (`plan.md`) — v2

- **Problem Statement ID:** SIH26262
- **Title:** Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System
- **Organization / Department:** AICTE / Regulation Bureau
- **Category & Theme:** Software | Smart Automation
- **Idea Submission Deadline:** 28 October 2026 (the portal showed **8/500** ideas for this PS when last checked — submit the PPT early, do not wait for the last day)
- **Solution Name: PRAVAH-AICTE** — *Platform for Regulatory Approval, Vigilance, Automation & Handbook Compliance*
- **Execution model: One single continuous build.** No phases, no phase gates. All workstreams start together, integrate continuously on one trunk, and finish together against one Definition of Done (Section 12).

> **How to read this document:** Sections 1–3 explain *what* and *why* (and map every line of the problem statement to a feature and a test). Sections 4–9 are the technical blueprint (architecture, workflow engine, modules, security, schema, APIs). Section 10–11 are the repo layout and seed data. Section 12 is the single-phase execution plan. Sections 13–14 are the demo script and the PPT content. Sections 15–16 are risks and acceptance tests. Appendix A lists everything changed from v1.

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1. Executive Summary & Vision

AICTE processes applications from technical institutions through multiple stages — scrutiny / re-scrutiny, Expert Visit Committee (EVC), SAC/SHC and other stages under the **Approval Process Handbook (APH)** — and issues approvals (EoA for existing institutions, LoA for new ones), some of them **subject to time-bound undertakings** (for example, a real 2026-27 SAC/SHC report recommends "EoA with undertaking for the recruitment" of faculty). Tracking application movement, stage-wise pendency, EVC scheduling, undertakings, compliance submissions and document/lease validity across **multiple Regional Offices** and Bureaus is today a manual, follow-up-heavy exercise.

Our solution: PRAVAH-AICTE is a **configurable, event-driven Regulatory Workflow & Compliance Automation Platform** with six integrated modules:

1. **A configurable APH Workflow Engine** — approval tracks are *data* (versioned JSON definitions), not code. Zero-touch stage routing, authorised revert/return with recorded reasons.
2. **An SLA & Pendency Command Center** — working-day SLA clocks with pause/resume, ageing heatmaps, bottleneck and workload analytics, role-based dashboards.
3. **An EVC Queue & Scheduler** — auto-queueing after scrutiny, mode-aware (Physical / Online / Hybrid) scheduling, conflict-of-interest check, due/overdue tracking.
4. **An Undertaking & Compliance Ledger** — every conditional approval becomes a tracked record with due dates, evidence upload, officer verification and breach escalation.
5. **A Document Validity Monitor** with **AI-assisted extraction (human-confirmed)** of lease/certificate dates, tiered renewal alerts and `COMPLIANCE_DEFICIENT` flagging.
6. **A Notification, Escalation & Reporting Engine** backed by a **tamper-evident, hash-chained audit trail** and a strict **Human-in-the-Loop (HITL)** guard so automation can never take a statutory decision.

Design principles (these are what make it the "best" submission, not just a complete one):

Principle	What it means in practice
Handbook-as-data	APH tracks, SLA days, thresholds, escalation chains, reminder offsets and document types are all configuration with versioning and effective dates. When APH changes, no deployment is needed.
Deadlines that cannot be missed	A *deterministic, idempotent sweeper* recomputes due actions from the database on every run (and on restart), so a crashed worker or a day of downtime still results in every alert being sent exactly once.
Automation assists, humans decide	The state machine *physically cannot* execute statutory transitions (approve / reject / show-cause) unless the actor is an authorised human who passed step-up authentication. Enforced in code and covered by tests.
Prove it, don't claim it	A live **Proof Panel** shows alert timeliness, missed-deadline count, audit-chain verification and SLA stats — mapped to the exact evaluation criteria in the problem statement.
Demo-safe by design	A virtual clock ("Time Travel") lets judges watch days of deadlines elapse in seconds; one command boots everything; a reset button restores the seed.

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2. Ground Truth: Facts Verified Against AICTE Sources (and What Must Still Be Verified)

Several details in v1 were assumptions. The table below corrects them and flags what to confirm in the official **APH 2024-25 to 2026-27** handbook before the final demo.

#	Fact	Evidence / status	How the plan handles it
1	AICTE has **more than four** Regional Offices — public AICTE documents list Western, Southern, South Western, South Central, Central, Eastern, North Western, Northern and North Eastern. Approval letters carry prefixes such as *South-West*, *South-Central*, *Southern*.	Verified from AICTE RO list / RTI page / EoA letters. Exact current list: **confirm**.	`Region` is a **lookup table**, not a 4-value enum. Seed with all nine; map-based HQ view is built from the table.
2	**SAC = Standing Appellate Committee** (considers appeals). Real committee reports are titled **"SAC/SHC"** with fields *Committee ID, SAC/SHC Status, Completion Date, EoA/LoA Recommended*.	Verified from AICTE 2026-27 SAC reports and the Dec-2025 corrigendum. **SHC** expansion: **confirm** in APH.	Modelled as **one committee stage** `SAC_SHC_REVIEW` with outcomes `RECOMMENDED`, `RECOMMENDED_WITH_UNDERTAKING`, `NOT_RECOMMENDED`. Labels are configurable. v1 wrongly showed `SHC_Hearing → SAC_Appeal` as two sequential stages.
3	**EVC is constituted by the Council** (corrigendum dated 2 Dec 2025, changing "Regional Officer"), and can be **Physical / Online / Hybrid**.	Verified (TeamLease RegTech summary of the corrigendum).	`EvcSchedule.mode`, `constitutedBy`; the RO's role is *queue visibility + escalation recipient*, not committee constitution. Config flag allows either model.
4	Approvals can be **with undertaking** (e.g. faculty recruitment).	Verified from a 2026-27 SAC/SHC report.	Undertaking ledger seeded with this exact real-world case type.
5	Permanent institution ID format looks like `1-46273286079`; EoA numbering like `South-West/1-46287070878/2026/EOA`.	Seen on public EoA letters.	Seed data and validators use this format.
6	Applications are submitted through the AICTE web portal (and the 2026-27 notice references NSWS).	Public notice. **Confirm** current integration surfaces.	Integration is via an **adapter layer** (Section 7.6) — mocked in the demo, replaceable in production.
7	Exact **stage SLA durations**, EVC completion windows, undertaking periods, lease validity norms (e.g. "30-year registered lease")	**Not verified** — pull from APH.	All durations are **configuration** with clearly marked `PLACEHOLDER` seed values. Never present placeholders to judges as official numbers; say "configured from APH".
8	Full list of tracked document types (lease deed, fire safety, building completion/occupancy, FDR/bank guarantee, etc.)	Partly from PS wording ("lease documents and other prescribed documents"). **Confirm** list.	`DocType` is a configurable table with per-type validity rules and reminder offsets.

> **Rule for the team:** anything in a PPT slide or the demo that states an *official* number (days, years, thresholds) must be traceable to an APH clause. Otherwise label it "configurable — set per APH".

3. Requirement Traceability Matrix (PS → Feature → Demo Proof → Test)

Evaluators score against the problem statement. This matrix guarantees nothing is missed and gives you a one-glance slide/appendix.

3.1 Functional requirements

PS Section	Requirement	Feature in PRAVAH-AICTE	Demo proof	Automated test
A	Auto-route through applicable stages (Only Scrutiny, Scrutiny+EVC, EVC+SAC/SHC, Only EVC, others)	Rule-based track assignment + versioned workflow definitions (Module A)	Submit 4 applications → 4 different tracks assigned	`workflow.tracks.spec` — every track, every transition
A	Auto-move to next stage on completion of preceding stage	Transition service with guards; `onEnter` actions	Officer clears scrutiny → file lands in EVC queue instantly	`workflow.autoadvance.spec`
A	Authorised return/revert with reasons recorded	`REVERT_FOR_DEFICIENCY` with structured reason codes + remarks; pauses officer SLA, starts institution response clock	Revert demo, audit entry, institution notification	`revert.spec` (reason mandatory, SLA pause/resume correct)
A	Automate routine communications & electronic case summaries	Template engine (EN/HI) + Case Summary PDF (draft, officer-finalised)	One-click case summary download	`casesummary.snapshot.spec`
B	Real-time consolidated dashboard (region, stage, institution, officer)	Command Center with live WebSocket updates	HQ dashboard filters	`dashboard.aggregates.spec`
B	Ageing, timelines, workload, bottlenecks	SLA % engine, workload distribution, bottleneck stage ranking	Bottleneck heatmap	`sla.calc.property.spec`
B	Alerts for approaching/exceeding timelines	Sweeper + escalation matrix	Time-travel demo	`sweeper.exactlyonce.spec`
B	Role-based dashboards	5 role views with scoped data access	Log in as each role	`rbac.scope.spec`
B	Complete digital audit trail	Hash-chained append-only log + verifier	Audit viewer "Verify chain ✓"	`audit.chain.spec`, `audit.tamper.spec`
C	Identify applications needing EVC, queue after preliminary processing	`EVC_QUEUE` on-enter action, priority scoring	Auto-jump to EVC queue	`evc.queue.spec`
C	Track EVC scheduling/completion (esp. new institutions)	EVC tracker with new-institution priority & deadlines	EVC calendar, overdue badge	`evc.deadlines.spec`
C	Reminders and escalation of delayed EVCs	T-5 / T-2 / T+1 style offsets (configurable)	Time-travel +15 days	`evc.escalation.spec`
D	Digital record of approvals with time-bound undertakings, conditions, evidence, due dates	Undertaking Ledger auto-created on `APPROVED_WITH_UNDERTAKING`	Undertaking list per institution	`undertaking.create.spec`
D	Notify institutions of upcoming deadlines; online submission of documents	Tiered alerts + secure evidence upload	Institution portal upload	`undertaking.alerts.spec`
D	Track submission & verification by authorised officials	Submit → Verify/Reject → Close workflow	Officer verification screen	`undertaking.verify.spec`

PS Section	Requirement	Feature in PRAVAH-AICTE	Demo proof	Automated test
D	Reminders & escalation when unfulfilled beyond timeline	`BREACHED_ESCALATED` + escalation chain	Fast-forward demo	`undertaking.breach.spec`
E	Track lease/document validity and expiry	Tracked-document register with `expiryDate`	Expiry calendar	`docs.expiry.spec`
E	Advance alerts, renewal submission, verification	90/60/30/15/7 (configurable) + renewal upload + AI-assisted verification	Lease renewal flow	`docs.renewal.spec`
E	Overdue alerts & escalation for expired/non-renewed	`EXPIRED` + `COMPLIANCE_DEFICIENT` + escalation	Expired lease escalated	`docs.expired.spec`
F	Configurable notifications for all event types	Template + rule configuration UI	Edit a reminder offset live	`notif.config.spec`
F	Auto-escalate overdue matters per timelines	Multi-level escalation matrix	Level 0/1/2 chain	`escalation.matrix.spec`
F	Consolidated reports (pendency, EVC status, undertakings, document validity, overdue, region-wise performance)	Reports module (6 reports, XLSX/PDF, scheduled)	Report download	`reports.spec`
F	Complete audit trail of alerts, submissions, verification, escalations, closure	Every one of those emits an audit event	Audit viewer filtered by case	`audit.coverage.spec`

3.2 Key requirements

Key requirement	How satisfied
Single integrated platform	One monorepo, one database, one audit chain across approval + compliance
Configurable rules/timelines aligned with APH	Versioned `WorkflowDefinition`, `SlaPolicy`, `EscalationRule`, `DocType` tables + admin UI
Secure storage, RBAC, confidentiality, traceable actions	Encrypted vault, malware scan, region-scoped data access, step-up auth, audit chain (Section 7)
Integration with existing AICTE systems where feasible	Adapter layer + OpenAPI + bulk import (Section 7.6)
Human oversight for regulatory decisions	Statutory-action guard in the state machine (Section 5.5)

3.3 Evaluation criteria → how we *prove* each one

Evaluation criterion (from PS)	Proof artifact in the prototype
Workflow accuracy	Golden-path suite over all tracks; transition table exported to the PPT; "illegal transition" attempts are rejected and logged
Timeliness of alerts	Proof Panel shows *scheduled time vs dispatched time* for every alert; target: dispatched within one sweep interval of the threshold
Reliability of deadline tracking	**365-day simulation run** over 5,000 synthetic records asserting **exactly one alert per tier per entity** and **zero missed deadlines**, including a simulated 24-hour worker outage with catch-up
Visibility of pending cases	HQ command center + saved filters ("Red/Critical", "EVC overdue", "Leases expiring ≤ 30 days")
Reduction in manual follow-up	Counter of "manual follow-ups avoided" (each auto-sent reminder/escalation that would have been a phone call/email) — framed as a *measurable pilot metric*, not a guaranteed percentage
Completeness of digital audit trail	Audit coverage test (every state-changing endpoint emits an event) + one-click chain verification

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4. System Architecture & Definitive Tech Stack

4.1 Architecture diagram (use on PPT Slide 3/4)

4.2 Architecture decisions (each with a one-line rationale — useful for judge Q&A)

#	Decision	Rationale
D1	**TypeScript pnpm monorepo** (`apps/web`, `apps/api`, `apps/worker`, `packages/*`)	One language, shared Zod schemas/types between UI and API, six people can work in parallel without contract drift.
D2	**NestJS API (Fastify adapter)** — *not* "NestJS or FastAPI"	v1 left this open and the folder structure assumed Next.js API routes. A definitive choice removes ambiguity; Next.js is UI only.
D3	**Workflow = XState v5 machines *generated from* versioned JSON definitions** + `json-rules-engine` for track assignment	Handbook-as-data. Machines are built at runtime from the active definition version.
D4	**Stages are string keys, not a Prisma enum**	A no-code configurator is impossible if stages are compiled into an enum (v1 inconsistency).
D5	**Deterministic sweeper + BullMQ for delivery** (not "one delayed job per deadline")	Delayed jobs are lost/duplicated on Redis flush or deploys and cannot follow a virtual clock. The sweeper derives due work from DB columns (`nextAlertAt`), is idempotent via unique `dedupeKey`, and catches up after downtime. BullMQ handles retry/backoff of *delivery* only.
D6	**Virtual `ClockService`** — every piece of code calls `clock.now()`, never `Date.now()`	Enables the Time-Travel demo and the 365-day simulation test without touching row data. Offset stored in `system_clock`; only active when `DEMO_MODE=true`.
D7	**Transactional outbox** for events (state change + event row in one DB transaction)	Guarantees no lost alerts/audit events if the process dies between "state changed" and "notify".
D8	**Business logic in a service layer, not DB triggers** (v1 proposed a DB trigger for EVC queueing)	Testable, observable and versioned with the workflow definition; triggers hide logic from reviewers.
D9	**Working-day SLA with holiday calendar and pause/resume**	Government timelines are working-day based and the clock must stop while the file is with the institution.
D10	**Per-stream hash chain + daily Merkle anchor** (not one global chain)	A single global chain serialises every write and races under concurrency; per-application chains scale, and a daily anchor row ties all stream heads together.
D11	**Document extraction = deterministic baseline + optional LLM, always human-confirmed**	The demo must work offline; the officer remains the decision-maker; avoids sending sensitive documents to external APIs in production.
D12	**Scoped Prisma client per request** (+ Postgres RLS as hardening)	Region/institution scoping enforced in one place rather than in every query.

4.3 Definitive tech stack

- **Frontend:** Next.js 15 (App Router, TypeScript), Tailwind CSS, `shadcn/ui`, TanStack Table v8, Recharts, a lightweight map (`react-simple-maps` / SVG India map by region), `next-intl` (English + Hindi), Socket.io-client for live toasts, `react-big-calendar` for EVC scheduling, `react-pdf` for in-browser PDF/lease preview with extracted-clause highlighting.
- **Backend:** Node 20, **NestJS 10 + Fastify**, Zod (shared validation), OpenAPI/Swagger auto-generated, `class-validator` only at the edges.
- **Workflow & Rules:** `xstate@5`, `json-rules-engine`, a custom `workingDays` utility (holiday-aware).
- **Database:** PostgreSQL 16, Prisma ORM, `pg_trgm` for search, advisory locks for audit chain writes, DB trigger *only* to block UPDATE/DELETE on `audit_logs`.
- **Queues & scheduling:** Redis 7 + **BullMQ** (delivery, retries, OCR/PDF jobs) + NestJS `@nestjs/schedule` for the sweeper tick.
- **Document vault & intelligence:** MinIO (S3-compatible, SSE encryption) / AWS S3; ClamAV for malware scan; `pdfjs-dist` (text layer) → `tesseract.js` (scanned PDFs/images) → optional **Gemini Flash** via a provider interface; regex + date-grammar baseline so extraction works with no network.
- **Reporting:** `Puppeteer` (Electronic Case Summary PDF), `exceljs` (XLSX exports), cron-scheduled weekly digests.
- **Notifications:** SMTP (Mailpit in dev, NIC/SES in prod), SMS gateway adapter (**DLT-registered templates** are mandatory in India — mocked and logged in the demo), In-App via WebSocket, WhatsApp Business API as an *optional* adapter (not required for the prototype).
- **Auth:** OIDC (Keycloak in dev as stand-in for AICTE SSO), JWT access + refresh, **TOTP step-up** for statutory actions.

- **Quality:** Vitest/Jest, Playwright (E2E that *is* the demo script), Testcontainers (Postgres/Redis), k6 (optional load test), ESLint + TypeScript strict, GitHub Actions CI.
- **Deployment:** Docker Compose (`web`, `api`, `worker`, `postgres`, `redis`, `minio`, `clamav`, `mailpit`, `keycloak`) — **one command boots everything**; cloud deploy on a single VM / Render / Railway for a live demo link.

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5. Workflow Engine, SLA Engine & Safeguard Design (the core IP)

5.1 Workflow definition format (stored in DB, versioned, effective-dated)

Each approval track is a JSON document. **All durations below are `PLACEHOLDER` values — replace from APH.**

```json
{
  "code": "SCRUTINY_PLUS_EVC",
  "version": 3,
  "academicYear": "2026-27",
  "effectiveFrom": "2026-11-01",
  "initial": "SUBMITTED",
  "stages": {
    "SUBMITTED":           { "kind": "auto",  "next": "SCRUTINY" },
    "SCRUTINY":            { "kind": "human", "roles": ["SCRUTINY_OFFICER"], "slaWorkingDays": 7,
                             "actions": { "CLEAR": "EVC_QUEUE", "REVERT": "DEFICIENCY_REVERTED", "RECOMMEND_REJECT":
    "DEFICIENCY_REVERTED": { "kind": "institution", "responseWorkingDays": 10, "pausesOfficerSla": true,
                             "actions": { "RESUBMIT": "RE_SCRUTINY" }, "onTimeout": "ESCALATE_TO_RO" },
    "RE_SCRUTINY":         { "kind": "human", "roles": ["SCRUTINY_OFFICER"], "slaWorkingDays": 5,
                             "actions": { "CLEAR": "EVC_QUEUE", "REVERT": "DEFICIENCY_REVERTED" } },
    "EVC_QUEUE":           { "kind": "queue", "slaWorkingDays": 5, "onEnter": ["CREATE_EVC_SCHEDULE"],
                             "actions": { "SCHEDULE": "EVC_SCHEDULED" } },
    "EVC_SCHEDULED":       { "kind": "human", "roles": ["EVC_MEMBER"], "slaWorkingDays": 15,
                             "actions": { "SUBMIT_REPORT": "COMPETENT_AUTHORITY_REVIEW", "RESCHEDULE": "EVC_SCHEDULE
    "COMPETENT_AUTHORITY_REVIEW": { "kind": "human", "statutory": true, "roles": ["REGIONAL_OFFICER", "BUREAU_HEAD"]
                             "actions": { "APPROVE": "APPROVED_FINAL", "APPROVE_WITH_UNDERTAKING": "APPROVED_WITH_UN
    "APPROVED_WITH_UNDERTAKING": { "kind": "terminal", "onEnter": ["CREATE_UNDERTAKING_RECORDS", "ISSUE_APPROVAL_DRA
    "APPROVED_FINAL":      { "kind": "terminal", "onEnter": ["ISSUE_APPROVAL_DRAFT", "SCHEDULE_DOC_WATCHERS"] },
    "REJECTED":            { "kind": "terminal" }
  }
}
```

- `statutory: true` on a stage or action marks it as a **statutory decision** → protected by the HITL guard (5.5).
- The four required families map to definitions: **ONLY_SCRUTINY**, **SCRUTINY_PLUS_EVC**, **EVC_PLUS_SAC_SHC** (EVC → `SAC_SHC_REVIEW` → competent authority), **ONLY_EVC**, plus an extensible `OTHER_COMBINATION` slot (the PS says "and other combinations prescribed under the APH"). A fifth/sixth track can be added from the admin UI **live in the demo**.
- **Versioning:** applications are pinned to the definition version active at submission (`workflowDefinitionId`). New versions apply to new applications; an admin can optionally migrate in-flight applications with an audited bulk action.
- **Definition validator** (runs on save): every stage reachable, no dead ends, every `actions` target exists, at least one terminal stage, statutory stages have roles, SLA > 0 for human stages. Invalid definitions cannot be activated.

5.2 Track assignment (rules engine)

```json
{
  "conditions": { "all": [
    { "fact": "applicationType", "operator": "equal", "value": "NEW_INSTITUTION" } ] },
  "event": { "type": "ASSIGN_TRACK", "params": { "track": "SCRUTINY_PLUS_EVC" } },
  "priority": 100
}
```

Facts available to rules: `applicationType`, `isNewInstitution`, `intakeDeltaPct`, `hasOpenDeficiency`, `previousApprovalYears`, `hasPendingUndertaking`, `regionCode`, `programLevel`. Rules are priority-ordered; the **rule that fired is stored on the application**

(`trackAssignmentReason`) so the officer — and the audit trail — can see *why* a track was chosen. Admin UI includes a **"dry-run" panel** (enter facts → see the track and path).

5.3 SLA engine

Let *W(a,b)* be the number of **working days** between *a* and *b* (weekends and the `HolidayCalendar` excluded; regional holidays supported), and *P* the total **paused** working time while the file sat with the institution.

$$\text{SLA Consumed (\%)} = \frac{W(\text{StageEntry},\ \text{clock.now}()) - P}{\text{Prescribed Stage Duration}} \times 100$$

One threshold set, used everywhere (v1 used 70% in one place and 75% in another):

State	SLA consumed	Dashboard colour	Automated action
`GREEN`	< 75%	Green	none
`AMBER`	75% – < 100%	Amber	**Level 0:** reminder to assigned officer (in-app + email)
`RED`	100% – < 125%	Red	**Level 1:** escalate to Regional Officer / Assistant Director
`CRITICAL`	⩾ 125%	Dark red	**Level 2:** escalate to Director, Regulation Bureau (HQ)
`PAUSED`	clock stopped	Grey/blue	institution response clock runs instead

- Thresholds (75/100/125) and the number of levels are rows in `EscalationRule`, not constants.
- **Pause/resume:** entering an `institution`-kind stage sets `slaPausedAt`; leaving it adds elapsed working time to `slaPausedTotalSec` and recomputes `stageSlaDeadline`. Property-based tests (fast-check) verify that for any sequence of pauses the deadline equals entry + duration + total paused working time.
- `nextAlertAt` is persisted on each entity = the earliest future threshold crossing; the sweeper only needs `WHERE nextAlertAt <= clock.now()` (indexed) — O(due items), not O(all items).

5.4 The deterministic sweeper (reliability of deadline tracking)

```
every 60s (and once on worker boot):
  now = clock.now()
  rows = SELECT entity FROM (applications, evc_schedules, undertakings, tracked_documents)
         WHERE nextAlertAt <= now AND status is active   -- indexed, LIMIT batch, FOR UPDATE SKIP LOCKED
  for each row:
     compute due tier(s) (could be several if the system was down)
     for each tier: INSERT notification(dedupeKey = "<entity>:<id>:<tierKey>")  ON CONFLICT DO NOTHING
     apply state change (SLA state / escalationLevel / BREACHED / EXPIRED) + audit event + outbox, same
transaction
     recompute nextAlertAt
  delivery workers pick PENDING notifications → send → mark SENT / retry with backoff / dead-letter
```

Properties (each is a test): **idempotent** (running twice never double-sends), **catch-up safe** (24h outage → on restart every missed tier is sent once, in order, flagged `isCatchUp=true`), **crash-safe** (state change, audit and outbox row commit atomically), **observable** (Proof Panel shows lag = `dispatchedAt - dueAt`).

5.5 Human-in-the-Loop (Regulatory Safeguard) — enforced, not just stated

1. Every action in a definition has a `statutory` flag (default false; `APPROVE`, `APPROVE_WITH_UNDERTAKING`, `REJECT`, `ISSUE_SHOW_CAUSE`, `CLOSE_UNDERTAKING_AS_BREACHED`, `SAC_SHC_OUTCOME` are true).
2. The state machine guard rejects a statutory action unless **all** hold: `actor.type === USER`, `actor.role ∈ stage.roles`, actor's region/scope matches, **step-up authentication** (fresh TOTP, ⩽ 5 min) verified, and a **mandatory remarks** field is present.
3. `actor.type === SYSTEM` can never trigger a statutory action — unit-tested with a "system tries to approve" test that must fail.
4. Every system-generated artefact (case summary, recommended track, extracted date, suggested SLA reason) carries a visible **"System-generated — for officer verification"** badge and records `generatedBy = SYSTEM` + the officer who accepted it (`acceptedById`).
5. Show-cause / penalty review is **only ever suggested** (a queue item for the Bureau), never auto-issued.
6. The `ActionSignature` table links each statutory audit entry to the step-up method (`TOTP_STEP_UP` in the prototype; `ESIGN` / `DSC` adapters for production).

5.6 Audit trail design (tamper-*evident*)

- **Streams:** one hash chain per `streamKey` (`APP:<applicationId>`, `INST:<institutionId>`, `SYS`). Within a stream, `seq` increases by exactly 1; writes take a `pg_advisory_xact_lock(hashtext(streamKey))`.
- **Canonical hash input** (JSON canonicalised with sorted keys, UTC ISO-8601 timestamps, so verification is reproducible):

$$\text{hash}_n = \text{SHA256}\big(\text{hash}_{n-1} \parallel \text{streamKey} \parallel \textit{seq}_n \parallel \text{actorType} \parallel \text{actorId} \parallel \text{actionType} \parallel \text{entity} \parallel \text{SHA256}(\text{payload}) \parallel \text{timestamp}\big)$$

- The first entry in a stream uses a fixed genesis value (`hash_0 = SHA256("PRAVAH-GENESIS|" + streamKey)`).
- **Database enforcement:** a trigger raises an exception on any `UPDATE` / `DELETE` of `audit_logs`; the application DB role has `INSERT, SELECT` only.
- **Daily anchor:** the worker computes a Merkle root over every stream head and stores it in `audit_anchors`; the anchor can additionally be emailed to the Bureau Head or notarised externally in production. This detects wholesale truncation of a stream, not just edits.
- **Verification:** `GET /audit/verify?stream=APP:<id>` recomputes the chain and returns `OK` or the first broken `seq`. The UI has a **"Verify chain"** button, and a **"Simulate tampering"** demo toggle (edits a row in a sandbox copy) to *show* detection.
- Wording: call it **tamper-evident** (v1 said "tamper-proof" — a database administrator can always alter data; what the design guarantees is that alteration is detected).

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6. Module-by-Module Implementation Blueprint (Modules A–F)

Module A — End-to-End Approval Workflow Automation

Capabilities

- **Intake & track assignment:** application arrives (portal adapter / bulk import / demo "new application" form) → rules engine assigns track → `ApplicationStageHistory` row opened → SLA deadline computed → first alert time set.
- **Auto-advance:** completing a human action moves the file to the next stage in the **same transaction** (state change + audit + outbox); stage `onEnter` actions run (create EVC schedule, create undertaking records, draft approval letter, schedule document watchers).
- **Authorised revert / return (deficiency loop):**
  - Structured **reason codes** (configurable list, e.g. *Document missing, Document illegible, Land/lease deficiency, Faculty deficiency, Infrastructure clarification, Fee/FDR issue, Other*) + free-text remarks + optional per-item **deficiency checklist** (`Deficiency` + `DeficiencyItem`).
  - Revert pauses the officer's SLA, starts the institution's response clock, notifies the institution immediately (in-app + email + SMS), and shows the checklist in the institution portal.
  - Institution uploads clarifications per item → file auto-routes to **Re-Scrutiny** (or the configured next stage) and the officer SLA resumes.
  - If the institution does not respond within the response window → reminder ladder, then escalation to the RO (never an automatic rejection — rejection is statutory).
- **Further verification / clarification:** an officer can send a *clarification request* (lighter than a deficiency revert) that does not change stage but opens a timed query thread.
- **Bulk & delegated actions:** reassign officer, bulk reassign on leave (audited), delegation with start/end date.
- **Electronic Case Summary** (see Module F-2): includes institution profile, track and path taken, stage timeline with durations, all deficiencies and responses, EVC findings, committee outcome, pending undertakings, document validity status, SLA performance — rendered as a **DRAFT** PDF with a content hash; the competent authority "finalises" it, which writes an audit event.
- **Standardised communications:** template library (EN/HI) — acknowledgement, deficiency notice, EVC intimation, undertaking reminder, expiry reminder, escalation notice, closure notice — with variable substitution and preview; all templates are versioned.

Key endpoints: `POST /applications`, `GET /applications/:id`, `POST /applications/:id/actions/:action`, `POST /applications/:id/revert`, `POST /applications/:id/resubmit`, `POST /applications/:id/reassign`, `GET /applications/:id/timeline`, `POST /applications/:id/case-summary`.

Edge cases handled: duplicate submission, resubmission after rejection (new application, linked history), withdrawal, definition version change mid-flight, officer on leave, region transfer, action attempted from a stale UI (optimistic-lock `version` field → 409).

Module B — Integrated Dashboard & Pendency Monitoring

Role-based views (each backed by scoped queries — a Regional Officer can never see another region unless granted):

Role	View	Highlights
Bureau Head / Bureau Officer (HQ)	**National Command Center**	Map coloured by region SLA health; funnel by stage; top bottleneck stages; officer workload distribution; "Red & Critical" feed; EVC overdue; undertakings breached; leases expired; region league table
Regional Officer	**Regional Desk**	Region queue; escalations needing intervention; EVC scheduling board; officer-wise workload with reassign; upcoming compliance deadlines for region institutions
Scrutiny Officer / EVC Member / SAC-SHC Member	**My Work**	Assigned files sorted by SLA urgency; keyboard-friendly action bar; deficiency modal; EVC calendar; conflict-of-interest warnings
Institution Admin	**Institution Portal**	Live pizza-tracker of the application; deficiency checklist; undertaking list with due countdown; document expiry calendar; notification inbox; downloadable acknowledgements
Auditor	**Audit Viewer** (read-only)	Chain verification, filters by case/actor/action, export
System Admin	**Policy Configurator**	Workflow definitions, SLA policies, escalation matrix, templates, holiday calendar, doc types

Analytics (all computed server-side, cached with short TTL, pushed by WebSocket on change):

- Ageing buckets (0–7, 8–15, 16–30, 30+ working days) by stage and region
- **Bottleneck index** per stage = mean SLA% × volume
- **Workload balance** = open files per officer vs. region median (Gini or max/median ratio)
- Throughput and median turnaround by track/application type
- **Forecast:** files predicted to breach in the next 7 days (based on current SLA% and remaining time) — shown as *advisory*
- Saved views and shareable filter URLs; CSV/XLSX export of any table

Performance targets (tested): dashboard aggregates p95 < 1 s on 50,000 synthetic applications (via indexed `stage`, `regionId`, `slaState`, `nextAlertAt`), live update latency < 2 s.

Module C — EVC Scheduling & Alerts

- **Auto-queueing:** `EVC_QUEUE.onEnter` creates an `EvcSchedule` (status `QUEUED`) in the application's region queue; the queue is ordered by a transparent **priority score** = new-institution weight + SLA% + days in queue + configured deadline proximity (score breakdown visible on hover — no black box).
- **Mode:** `PHYSICAL | ONLINE | HYBRID` (per the Dec-2025 corrigendum); the online mode skips travel logistics fields.
- **Committee composition:** `EvcCommitteeMember` rows (lead + members), configurable minimum size, **conflict-of-interest check** (member's institution/state vs. applicant; prior association flag) before scheduling is confirmed.
- **Scheduling calendar:** drag-to-schedule, reschedule with mandatory reason (history kept, `rescheduleCount` shown), clash detection for members.
- **Deadlines:** `prescribedDeadline` (visit to be completed) and `reportDueAt` (report to be submitted) tracked separately — new institutions use a stricter window (configurable).
- **Reminders/escalation (offsets configurable per rule):** unscheduled for N days after queueing → reminder to RO; visit due in 5 and 2 working days → reminder to committee lead; report overdue at +1 day → RO; +3 days → Bureau.
- **Report intake:** structured EVC report upload (score/observations + attachments) → triggers auto-advance to the next stage.

Module D — Time-Bound Undertakings & Institutional Compliance

- **Ledger creation:** on `APPROVE_WITH_UNDERTAKING` (or on SAC/SHC outcome `RECOMMENDED_WITH_UNDERTAKING`) the officer selects/creates conditions from a **condition library** (e.g. *Recruit and upload faculty details*, *Barrier-Free Environment certificate*, *Fire safety certificate*, *Anti-ragging committee constitution*, *Display of information on website*), each with a due date, evidence requirements and verifying role. One `Undertaking` per condition.
- **Lifecycle:** `PENDING_SUBMISSION → SUBMITTED_UNDER_REVIEW → VERIFIED_COMPLIANT | DEFICIENT_REVERTED → (resubmit) …` and `OVERDUE_ESCALATED` / `BREACHED` when past due.
- **Reminder ladder (configurable):** T-30, T-15, T-7, T-3 working/calendar days (calendar-day basis by default for institution-facing deadlines — set per rule), due day, then overdue ladder +1, +7, +15.

- **Evidence submission:** drag-and-drop (PDF / images, geo-tagged photo support — EXIF GPS/time shown to the verifier), malware scan, SHA-256 fingerprint, versioned.
- **Verification:** assigned official clicks `VERIFY_AND_CLOSE` or `REJECT_RESUBMIT` (reason mandatory). Closing is recorded with an `ActionSignature`.
- **Extension requests:** the institution can request an extension with justification; approval is a statutory-flagged action by the competent authority (never auto-granted); extension history is retained.
- **Breach handling:** `current_date > due_date` and not verified → `BREACHED_ESCALATED`, escalation to RO then Bureau, **and a "Show-cause review" queue item is created for human decision** — the system never issues penalties.
- **Institution compliance score (advisory):** weighted view of on-time undertakings, document validity and response times, used as a **watchlist** signal for officers, with the formula visible.

Module E — Lease & Document Validity Monitoring

- **Configurable document types** (table `DocType`): Registered Lease Deed, Fire Safety Certificate, Building Completion/Occupancy Certificate, Bank Guarantee/FDR, plus any prescribed document the APH specifies. Each type has: whether expiry applies, minimum validity rule (e.g. lease min-years check), reminder offsets, which stages require it.
- **Upload & extraction pipeline:** upload → malware scan → SHA-256 → text layer (`pdfjs-dist`) → OCR fallback (`tesseract.js`) → **extractor** returns `{documentNumber, issueDate, expiryDate/leaseTerm, registrationDetails, stampDuty}` with **per-field confidence** and the **source text span + page** → UI shows the PDF with the detected clause highlighted.
- **Human confirmation is mandatory:** extracted values pre-fill a form; the verifying officer confirms/edits; the final values (not the AI's) drive deadlines. Low-confidence fields are flagged amber and cannot be auto-accepted.
- **Cross-checks (advisory flags):** expiry before issue date; lease term shorter than the required minimum; document number format; date mismatch between form entry and document; file hash already used by another institution (possible reuse) — each shown as a flag for the officer, never an automatic rejection.
- **Lease-term logic:** if the lease is expressed as a term ("30 years from registration"), expiry is computed and the computation is shown.
- **Reminder ladder:** 90, 60, 30, 15, 7 days before expiry (configurable per DocType), expiry day, then overdue +1, +7, +30.
- **Renewal workflow:** institution uploads renewed document → new `FileVersion` → verification → previous version archived, expiry updated, alert ladder reset, overdue flags cleared, audit entry.
- **On expiry without renewal:** status `EXPIRED`, institution flagged `COMPLIANCE_DEFICIENT` (visible in the next approval cycle's scrutiny checklist), escalation to RO/Bureau.
- **Institution document calendar:** month view with colour-coded expiries; downloadable `.ics` feed so the institution's own calendar also reminds them.

Module F — Notifications, Escalation, Reporting & Audit

F-1 Notification engine

- **Event catalogue:** application received, stage assigned, SLA amber/red/critical, deficiency raised, response due/overdue, EVC queued/scheduled/due/overdue, undertaking T-x/overdue/breached, document T-x/expired/renewed, escalation raised/acknowledged/closed, report ready.
- **Rules (configurable in UI):** event → recipients (role/officer/institution) → channels → template → timing offsets → quiet hours → de-duplication window.
- **Channels:** In-App (WebSocket + inbox), Email, SMS (adapter), WhatsApp (optional adapter). Templates in **English and Hindi**; institution preferred language respected.
- **Delivery guarantees:** `dedupeKey` unique → at-most-once creation; BullMQ retries with exponential backoff; dead-letter queue visible to admins; delivery status and `readAt` stored (so "was this alert actually acknowledged?" is answerable).
- **Escalation acknowledgement:** an escalation stays "open" and re-notifies at the next level until a human **acknowledges** it with a remark — the chain doesn't just fire-and-forget.
- **Digest mode:** daily digest per officer (what's due today / this week) to reduce alert fatigue.

F-2 Reports (all with XLSX + PDF export, filters, and optional weekly scheduled email)

#	Report	Key columns / charts
R1	Application pendency	Region × stage × ageing bucket, officer-wise, SLA state
R2	EVC status	Queued / scheduled / completed / overdue, new-institution compliance %, mode split
R3	Undertaking compliance	Due / submitted / verified / breached by region & condition type
R4	Document validity	Expiring ⩽ 90 days, expired, renewal-in-progress, by DocType & region
R5	Overdue & escalations	Level-wise open escalations, time-to-acknowledge, repeat offenders
R6	Region-wise processing performance	Median turnaround, SLA adherence %, throughput, workload balance

F-3 Audit trail — see Section 5.6. Audit events are emitted for: stage transitions, reverts/resubmissions, document uploads/verifications, undertaking changes, alert dispatches, escalations & acknowledgements, configuration changes (who changed which SLA from what to what), logins to privileged roles, exports.

---

7. Cross-Cutting Concerns

7.1 Security architecture

- **AuthN/Z:** OIDC SSO (adapter), short-lived JWT + refresh rotation, RBAC + **scope** (region / institution / assigned-only), TOTP step-up for statutory actions, session timeout, concurrent-session alerts for privileged roles.
- **Data scoping:** scoped Prisma client injects region/institution filters; Postgres **RLS** policies as defence-in-depth in production; automated test that iterates every role × every endpoint and asserts no cross-scope leakage.
- **Document vault:** SSE encryption at rest, TLS in transit, **pre-signed URLs with ⩽ 5-minute expiry**, no public buckets, downloads **watermarked** with officer ID + timestamp, file-type allow-list by magic bytes, size limits, ClamAV scan with quarantine, SHA-256 fingerprinting.
- **Application hardening:** OWASP ASVS L2 checklist, Helmet/CSP, CSRF protection for cookie flows, rate limiting (Redis), input validation (Zod) at every boundary, parameterised queries only (Prisma), secrets via env/secret manager, dependency audit in CI, SBOM generation.
- **PII minimisation & retention:** store only data needed for the workflow; configurable retention; export/erasure workflow aligned with the **Digital Personal Data Protection Act, 2023** principles; no PII in logs.
- **Hosting note for the PPT:** production deployment on a MeitY-empanelled / NIC cloud with India data residency, CERT-In incident reporting readiness, and a pre-go-live security audit (e.g. STQC/CERT-In empanelled auditor). State this as the *deployment path*, not as something already done.

7.2 Accessibility, language & usability (government-grade polish)

- **WCAG 2.1 AA** and **GIGW 3.0** alignment: keyboard navigation, focus states, colour-contrast-safe SLA palette (never colour alone — icon + text label on every badge), screen-reader labels, reduced-motion support.
- **Bilingual UI & notifications:** English + Hindi (`next-intl`), extensible to regional languages.
- **Low-bandwidth resilience** for institution users: lightweight pages, resumable uploads, clear upload progress and retry.
- **Officer ergonomics:** keyboard shortcuts for clear/revert, saved filters, bulk actions with confirmation, undo for non-statutory changes.

7.3 Data & AI governance

- AI is used for **extraction and summarisation assistance only** — never for decisions.
- Every AI/OCR output stores: model/provider, version, confidence, source span, and the human who accepted or overrode it (the override rate is a quality metric on the Proof Panel).
- **Provider interface** (`DocumentIntelligenceProvider`) with three implementations: `RegexBaseline` (offline, deterministic — used in the demo by default), `TesseractOCR` , and `GeminiFlash` (optional). Production recommendation: on-premise / sovereign model or redaction before any external call; **demo documents are synthetic**.
- Prompt-injection safe: documents are treated as data, extraction output is schema-validated (Zod) and never executed.

7.4 Observability & operations

- Structured JSON logs with correlation IDs; OpenTelemetry traces; Prometheus metrics (sweeper lag, queue depth, alert latency, delivery failure rate); Grafana-style **Ops page** in the admin UI; health/readiness endpoints; sweeper heartbeat alert ("sweeper hasn't run in 3 minutes" raises an admin alert).
- Backups: nightly `pg_dump` + object-store versioning; documented RPO/RTO targets; restore drill noted in the runbook.

7.5 Configuration administration (the "No-Code APH Policy Configurator")

- Screens: **Workflow Designer** (stage graph editor with validation), **SLA Matrix** (stage × application-type grid), **Escalation Matrix** (level × threshold × role × channel), **Reminder Offsets**, **Template Editor** (EN/HI + preview), **Holiday Calendar**, **Document Types**, **Reason Codes**.
- Every change is versioned, effective-dated, requires `SYSTEM_ADMIN` + step-up, shows a **diff** against the active version, supports **rollback**, and writes a configuration audit event (who/what/old/new).
- A **dry-run simulator** previews the effect of a change on current open applications ("this SLA change would turn 14 files amber today").

7.6 Integration with existing AICTE systems (adapter layer)

All external dependencies sit behind interfaces with a **mock implementation for the demo** and a documented contract for production:

Adapter	Purpose	Prototype implementation	Production path
`AicteApprovalPortalAdapter`	Pull new/updated applications, push status back	Fixture-based mock + **CSV/XLSX bulk import** + webhook receiver	REST/webhook integration with the AICTE approval portal
`IdentityAdapter`	SSO / user directory	Keycloak (OIDC)	AICTE SSO / NIC identity
`MessagingAdapter`	Email, SMS, WhatsApp	Mailpit + logged SMS	NIC/SES SMTP, DLT-registered SMS gateway, WhatsApp Business API
`ESignAdapter`	Digital signature for statutory actions	TOTP step-up	eSign / DSC token
`DocumentSourceAdapter`	Fetch certificates directly	Upload only	DigiLocker / issuing-authority APIs where available

- **OpenAPI 3.1 spec** is published from the API; **webhooks** (signed with HMAC) let other AICTE systems subscribe to events like `application.stage_changed`, `undertaking.breached`, `document.expired`.
- **Migration path:** a one-time bulk-import template (Excel) lets a Regional Office onboard its existing pendency spreadsheet on day one — a strong adoption argument.

---

8. Complete Database Schema (`prisma/schema.prisma`)

Changes from v1: `Region` is a table (not an enum); stages are **string keys** (config-driven); added `WorkflowDefinition`, `RoutingRule`, `SlaPolicy`, `EscalationRule`, `HolidayCalendar`, `ApplicationStageHistory`, `Deficiency(+Item)`, `EvcCommitteeMember`, `FileVersion`, `Notification(+Template/Rule)`, `CaseSummary`, `OutboxEvent`, `ActionSignature`, `AuditAnchor`, `SystemClock`; SLA pause fields; `nextAlertAt` indexes for the sweeper; per-stream audit chain; optimistic-lock `version`. Run `npx prisma validate && npx prisma migrate dev` after pasting.

prisma

```
datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

// ──────────── Enums ────────────
enum Role {
  INSTITUTION_ADMIN
  SCRUTINY_OFFICER
  EVC_MEMBER
  SAC_SHC_MEMBER
  REGIONAL_OFFICER
  BUREAU_OFFICER
  BUREAU_HEAD
  AUDITOR
  SYSTEM_ADMIN
}

enum ActorType { USER SYSTEM }

enum ApplicationType {
  NEW_INSTITUTION
  EXTENSION_OF_APPROVAL
  INCREASE_IN_INTAKE
  NEW_COURSE
  NEW_PROGRAM_OR_LEVEL
  CHANGE_OF_SITE
  CLOSURE
  OTHER
}

enum ApplicationStatus {
  IN_PROGRESS
  WITH_INSTITUTION        // reverted / awaiting clarification (officer SLA paused)
  APPROVED
  APPROVED_WITH_UNDERTAKING
  REJECTED
  WITHDRAWN
}

enum SlaState { GREEN AMBER RED CRITICAL PAUSED }

enum EvcMode { PHYSICAL ONLINE HYBRID }

enum EvcStatus { QUEUED SCHEDULED COMPLETED OVERDUE_ESCALATED CANCELLED }

enum ComplianceStatus {
  PENDING_SUBMISSION
  SUBMITTED_UNDER_REVIEW
  VERIFIED_COMPLIANT
  DEFICIENT_REVERTED
  OVERDUE_ESCALATED
  BREACHED_ESCALATED
  EXTENSION_REQUESTED
  CLOSED
}

enum DocStatus { VALID EXPIRING EXPIRED RENEWAL_SUBMITTED RENEWED }
```

```
enum NotificationChannel { IN_APP EMAIL SMS WHATSAPP }

enum NotificationStatus { PENDING SENT FAILED READ DEAD_LETTER }

enum SignatureMethod { TOTP_STEP_UP ESIGN DSC }

enum ScanStatus { PENDING CLEAN INFECTED FAILED }

enum SummaryStatus { DRAFT FINALISED }

// ──────────── Organisation & identity ────────────
model Region {
  id           String            @id @default(uuid())
  code         String            @unique   // e.g. SRO, SCRO, SWRO, WRO, ERO, NRO, NWRO, CRO, NERO (confirm list)
  name         String
  users        User[]
  institutions Institution[]
  applications Application[]
  holidays     HolidayCalendar[]
}

model User {
  id             String       @id @default(uuid())
  name           String
  email          String       @unique
  role           Role
  designation    String?
  regionId       String?
  region         Region?      @relation(fields: [regionId], references: [id])
  institutionId  String?
  institution    Institution? @relation(fields: [institutionId], references: [id])
  preferredLang  String       @default("en")   // "en" | "hi"
  totpSecretEnc  String?                       // encrypted TOTP seed for step-up
  isActive       Boolean      @default(true)
  delegatedToId  String?                       // leave delegation
  delegationEnds DateTime?
  createdAt      DateTime     @default(now())

  assignedApplications Application[]         @relation("AssignedOfficer")
  committeeMemberships EvcCommitteeMember[]
  notifications        Notification[]
}

model Institution {
  id               String         @id @default(uuid())
  aictePermanentId String         @unique   // e.g. "1-46273286079"
  name             String
  state            String
  district         String?
  institutionType  String                    // "Engineering", "Pharmacy", "Polytechnic", ...
  regionId         String
  region           Region         @relation(fields: [regionId], references: [id])
  complianceHealth String         @default("GOOD")  // GOOD | WATCHLIST | COMPLIANCE_DEFICIENT
  complianceScore  Float?                           // advisory, formula visible in UI
  users            User[]
  applications     Application[]
  undertakings     Undertaking[]
  documents        TrackedDocument[]
}

model HolidayCalendar {
  id       String   @id @default(uuid())
```

```
  date      DateTime @db.Date
  label     String
  regionId  String?              // null = national
  region    Region?  @relation(fields: [regionId], references: [id])

  @@unique([date, regionId])
}

// ──────────── Configuration (Handbook-as-data) ────────────
model WorkflowDefinition {
  id             String   @id @default(uuid())
  code           String                        // ONLY_SCRUTINY | SCRUTINY_PLUS_EVC | EVC_PLUS_SAC_SHC | ONLY_EVC | c
  version        Int
  academicYear   String
  name           String
  definition     Json                          // see Section 5.1
  isActive       Boolean  @default(false)
  effectiveFrom  DateTime
  supersededAt   DateTime?
  createdById    String
  createdAt      DateTime @default(now())
  applications   Application[]
  slaPolicies    SlaPolicy[]

  @@unique([code, version])
}

model RoutingRule {
  id         String  @id @default(uuid())
  priority   Int
  name       String
  conditions Json                             // json-rules-engine conditions
  trackCode  String
  isActive   Boolean @default(true)
}

model SlaPolicy {
  id                 String             @id @default(uuid())
  workflowDefinitionId String
  workflowDefinition  WorkflowDefinition  @relation(fields: [workflowDefinitionId], references: [id])
  stageKey           String
  applicationType    ApplicationType?     // null = all types
  durationWorkingDays Int                // PLACEHOLDER values until APH is mapped
  warnPct            Int                 @default(75)
  breachPct          Int                 @default(100)
  criticalPct        Int                 @default(125)

  @@unique([workflowDefinitionId, stageKey, applicationType])
}

model EscalationRule {
  id          String  @id @default(uuid())
  scope       String                       // APPLICATION_STAGE | EVC | UNDERTAKING | DOCUMENT | DEFICIENCY_RESPONSE
  level       Int
  triggerType String                       // PCT_OF_SLA | DAYS_BEFORE_DUE | DAYS_AFTER_DUE
  triggerValue Int
  targetRole  Role
  channels    NotificationChannel[]
  templateCode String
  isActive    Boolean @default(true)

  @@unique([scope, level, triggerType, triggerValue])
}
```

```
model DocType {
  code            String  @id              // LEASE_DEED | FIRE_SAFETY | OCCUPANCY_CERT | FDR | ...
  label           String
  hasExpiry       Boolean @default(true)
  minValidityYears Int?                    // e.g. lease minimum term, from APH
  reminderOffsets Int[]                    // days before expiry, e.g. {90,60,30,15,7}
  requiredAtStages String[]
}

model ReasonCode {
  code     String  @id
  label    String
  category String                          // DEFICIENCY | EXTENSION | RESCHEDULE | REJECT_RESUBMIT
  isActive Boolean @default(true)
}

// ──────────── Core workflow ────────────
model Application {
  id                   String            @id @default(uuid())
  applicationNumber    String            @unique      // e.g. "APH2026-S-00412"
  institutionId        String
  institution          Institution       @relation(fields: [institutionId], references: [id])
  regionId             String
  region               Region            @relation(fields: [regionId], references: [id])
  academicYear         String
  applicationType      ApplicationType
  isNewInstitution     Boolean           @default(false)
  workflowDefinitionId String
  workflowDefinition   WorkflowDefinition @relation(fields: [workflowDefinitionId], references: [id])
  trackAssignmentReason String?                        // which routing rule fired and why

  currentStage         String                          // stage key from the definition
  status               ApplicationStatus  @default(IN_PROGRESS)
  stageEnteredAt       DateTime
  stageSlaDeadline     DateTime?
  slaPausedAt          DateTime?
  slaPausedTotalSec    Int                @default(0)
  slaState             SlaState           @default(GREEN)
  escalationLevel      Int                @default(0)
  nextAlertAt          DateTime?                       // sweeper index
  assignedOfficerId    String?
  assignedOfficer      User?              @relation("AssignedOfficer", fields: [assignedOfficerId], references: [id]

  version              Int                @default(1)  // optimistic locking
  createdAt            DateTime           @default(now())
  updatedAt            DateTime           @updatedAt

  stageHistory         ApplicationStageHistory[]
  deficiencies         Deficiency[]
  evcSchedules         EvcSchedule[]
  undertakings         Undertaking[]
  caseSummaries        CaseSummary[]
  auditLogs            AuditLog[]

  @@index([regionId, currentStage, slaState])
  @@index([nextAlertAt])
  @@index([assignedOfficerId, slaState])
}

model ApplicationStageHistory {
  id            String      @id @default(uuid())
  applicationId String
```

```
  application   Application @relation(fields: [applicationId], references: [id])
  stageKey      String
  enteredAt     DateTime
  exitedAt      DateTime?
  slaDeadline   DateTime?
  pausedTotalSec Int        @default(0)
  actorId       String?
  actionTaken   String?                          // CLEAR | REVERT | ...
  slaStateAtExit SlaState?

  @@index([applicationId, enteredAt])
}

model Deficiency {
  id            String          @id @default(uuid())
  applicationId String
  application   Application     @relation(fields: [applicationId], references: [id])
  raisedById    String
  reasonCode    String
  remarks       String
  raisedAt      DateTime
  responseDueAt DateTime
  respondedAt   DateTime?
  status        String          @default("OPEN")   // OPEN | RESPONDED | ACCEPTED | RE_RAISED
  nextAlertAt   DateTime?
  items         DeficiencyItem[]
}

model DeficiencyItem {
  id           String        @id @default(uuid())
  deficiencyId String
  deficiency   Deficiency    @relation(fields: [deficiencyId], references: [id])
  description  String
  resolved     Boolean       @default(false)
  responses    FileVersion[]
}

// ─────────────── EVC ──────────────
model EvcSchedule {
  id                String              @id @default(uuid())
  applicationId     String
  application       Application         @relation(fields: [applicationId], references: [id])
  mode              EvcMode             @default(PHYSICAL)
  constitutedBy     String              @default("COUNCIL")   // per Dec-2025 corrigendum; configurable
  queuedAt          DateTime
  priorityScore     Float               @default(0)
  priorityBreakdown Json?                                     // transparent score explanation
  scheduledStart    DateTime?
  scheduledEnd      DateTime?
  prescribedDeadline DateTime
  reportDueAt       DateTime?
  reportSubmittedAt DateTime?
  rescheduleCount   Int                 @default(0)
  status            EvcStatus           @default(QUEUED)
  escalationLevel   Int                 @default(0)
  nextAlertAt       DateTime?
  members           EvcCommitteeMember[]

  @@index([status, nextAlertAt])
}

model EvcCommitteeMember {
  id               String      @id @default(uuid())
```

```
  evcScheduleId     String
  evcSchedule       EvcSchedule @relation(fields: [evcScheduleId], references: [id])
  userId            String
  user              User        @relation(fields: [userId], references: [id])
  isLead            Boolean     @default(false)
  conflictChecked   Boolean     @default(false)
  conflictFlag      Boolean     @default(false)

  @@unique([evcScheduleId, userId])
}

// ──────────── Compliance ────────────
model Undertaking {
  id              String           @id @default(uuid())
  institutionId   String
  institution     Institution      @relation(fields: [institutionId], references: [id])
  applicationId   String
  application     Application      @relation(fields: [applicationId], references: [id])
  conditionCode   String                      // from condition library
  conditionTitle  String
  description     String
  issuedAt        DateTime
  dueDate         DateTime
  originalDueDate DateTime
  extensionCount  Int              @default(0)
  status          ComplianceStatus @default(PENDING_SUBMISSION)
  verifierRole    Role             @default(REGIONAL_OFFICER)
  verifiedById    String?
  verifiedAt      DateTime?
  escalationLevel Int              @default(0)
  nextAlertAt     DateTime?
  evidence        FileVersion[]
  version         Int              @default(1)

  @@index([status, nextAlertAt])
  @@index([institutionId, dueDate])
}

model TrackedDocument {
  id               String        @id @default(uuid())
  institutionId    String
  institution      Institution   @relation(fields: [institutionId], references: [id])
  docTypeCode      String
  documentNumber   String?
  issuingAuthority String?
  issueDate        DateTime?
  expiryDate       DateTime?
  status           DocStatus     @default(VALID)
  currentVersionId String?
  escalationLevel  Int           @default(0)
  nextAlertAt      DateTime?
  versions         FileVersion[]

  @@index([status, nextAlertAt])
  @@index([institutionId, expiryDate])
}

model FileVersion {
  id                 String           @id @default(uuid())
  versionNo          Int              @default(1)
  storageKey         String
  fileName           String
  mimeType           String
```

```
  sizeBytes           Int
  sha256              String
  uploadedById        String
  uploadedAt          DateTime         @default(now())
  scanStatus          ScanStatus       @default(PENDING)
  extractionProvider  String?                              // RegexBaseline | TesseractOCR | GeminiFlash
  extractionJson      Json?                                // fields, confidence, source spans
  extractionConfidence Float?
  verifiedById        String?
  verifiedAt          DateTime?
  overriddenFields    String[]                             // fields the officer changed (quality metric)

  trackedDocumentId   String?
  trackedDocument     TrackedDocument? @relation(fields: [trackedDocumentId], references: [id])
  undertakingId       String?
  undertaking         Undertaking?     @relation(fields: [undertakingId], references: [id])
  deficiencyItemId    String?
  deficiencyItem      DeficiencyItem?  @relation(fields: [deficiencyItemId], references: [id])

  @@index([sha256])
}

// ──────────── Notifications ────────────
model NotificationTemplate {
  code     String
  channel  NotificationChannel
  lang     String
  subject  String?
  body     String
  version  Int                @default(1)

  @@id([code, channel, lang])
}

model Notification {
  id           String              @id @default(uuid())
  dedupeKey    String              @unique        // "<entity>:<id>:<tierKey>" → exactly-once creation
  recipientId  String
  recipient    User                @relation(fields: [recipientId], references: [id])
  channel      NotificationChannel
  templateCode String
  payload      Json
  relatedType  String
  relatedId    String
  dueAt        DateTime                           // when it SHOULD have been sent
  status       NotificationStatus  @default(PENDING)
  attempts     Int                 @default(0)
  dispatchedAt DateTime?                         // lag = dispatchedAt - dueAt (Proof Panel)
  readAt       DateTime?
  acknowledgedAt DateTime?
  isCatchUp    Boolean             @default(false)
  error        String?

  @@index([status, dueAt])
  @@index([recipientId, readAt])
}

model OutboxEvent {
  id          String   @id @default(uuid())
  type        String
  payload     Json
  createdAt   DateTime @default(now())
  processedAt DateTime?
```

```
  @@index([processedAt, createdAt])
}

// ──────────── Reports & summaries ────────────
model CaseSummary {
  id            String        @id @default(uuid())
  applicationId String
  application   Application   @relation(fields: [applicationId], references: [id])
  version       Int
  status        SummaryStatus @default(DRAFT)
  storageKey    String
  contentHash   String
  generatedAt   DateTime      @default(now())
  generatedBy   ActorType     @default(SYSTEM)
  acceptedById  String?                          // officer who finalised it
  finalisedAt   DateTime?

  @@unique([applicationId, version])
}

// ──────────── Audit (tamper-evident) ────────────
model AuditLog {
  id            String       @id @default(uuid())
  streamKey     String                           // APP:<id> | INST:<id> | SYS
  seq           Int
  applicationId String?
  application   Application? @relation(fields: [applicationId], references: [id])
  actorType     ActorType
  actorId       String?
  actorRole     Role?
  actionType    String                           // STAGE_TRANSITION | DEFICIENCY_REVERT | SLA_ESCALATION | ALERT_
  entityType    String
  entityId      String
  previousState String?
  newState      String?
  remarks       String?
  payload       Json
  payloadHash   String
  prevHash      String
  hash          String       @unique
  createdAt     DateTime     @default(now())
  signature     ActionSignature?

  @@unique([streamKey, seq])
  @@index([applicationId, createdAt])
}

model ActionSignature {
  id         String          @id @default(uuid())
  auditLogId String          @unique
  auditLog   AuditLog        @relation(fields: [auditLogId], references: [id])
  userId     String
  method     SignatureMethod
  signedAt   DateTime
  challengeRef String?
}

model AuditAnchor {
  id          String   @id @default(uuid())
  anchorDate  DateTime @unique @db.Date
  rootHash    String
  streamCount Int
```

```
  createdAt   DateTime @default(now())
}

// ──────────── Demo support ────────────
model SystemClock {
  id            Int     @id @default(1)
  offsetSeconds Int     @default(0)            // virtual time offset (DEMO_MODE only)
  demoMode      Boolean @default(false)
  updatedAt     DateTime @updatedAt
}
```

SQL to add in the first migration (audit immutability):

sql

```
CREATE OR REPLACE FUNCTION audit_logs_immutable() RETURNS trigger AS $$
BEGIN RAISE EXCEPTION 'audit_logs is append-only'; END; $$ LANGUAGE plpgsql;
CREATE TRIGGER audit_logs_no_update BEFORE UPDATE OR DELETE ON "AuditLog"
  FOR EACH ROW EXECUTE FUNCTION audit_logs_immutable();
-- App role: GRANT INSERT, SELECT ON "AuditLog" TO pravah_app;  (no UPDATE/DELETE)
```

---

9. API Conventions

- **REST + OpenAPI 3.1**, versioned `/api/v1`, JSON, Zod-validated; `Idempotency-Key` header supported on all `POST` actions; errors follow RFC 7807 (`application/problem+json`).
- **Optimistic concurrency:** state-changing calls include the entity `version`; stale writes return `409`.
- **Pagination:** cursor-based; **filtering/sorting** via whitelisted query params so saved dashboard views are shareable URLs.
- **Realtime:** WebSocket namespaces `/ws/alerts` (per-user), `/ws/dashboard` (per-scope) emit `application.updated`, `sla.state_changed`, `escalation.raised`, `notification.created`.
- **Webhooks (outbound):** HMAC-signed, retried with backoff; event types listed in Section 7.6.
- **Demo-only endpoints** (guarded by `DEMO_MODE=true`): `POST /demo/advance-time {days}` (moves the virtual clock and triggers an immediate sweep), `POST /demo/reset` (restore seed snapshot), `POST /demo/simulate-tamper` (sandboxed), `GET /demo/proof-panel`.

---

10. Monorepo Folder Structure

text

```
sih26262-pravah-aicte/
├── docker-compose.yml            # web, api, worker, postgres, redis, minio, clamav, mailpit, keycloak
├── .env.example
├── Makefile                      # make up | make seed | make demo | make test | make simulate
├── apps/
│   ├── web/                      # Next.js 15 (UI only)
│   │   └── src/app/
│   │       ├── (auth)/login/
│   │       ├── bureau-hq/        # Command Center, region map, bottlenecks, escalation feed
│   │       ├── regional-desk/    # RO queue, EVC board, workload & reassign
│   │       ├── my-work/          # Scrutiny / EVC / SAC-SHC officer inbox & action bar
│   │       ├── applications/[id] # Timeline, deficiencies, case summary, audit tab
│   │       ├── evc/              # Queue, calendar, committee, report intake
│   │       ├── compliance/       # Undertakings + document validity monitor
│   │       ├── institution/      # Pizza-tracker, deficiency responses, uploads, expiry calendar
│   │       ├── reports/          # R1–R6 + scheduler
│   │       ├── admin/            # Policy Configurator (workflow designer, SLA, escalation, templates, holidays)
│   │       ├── audit/            # Hash-chain viewer + Verify + tamper demo
│   │       └── proof-panel/      # Evaluation-criteria metrics (judge page)
│   ├── api/                      # NestJS
│   │   └── src/
│   │       ├── workflow/         # Module A: definition loader/validator, machine factory, transition service, guar
│   │       ├── sla/              # Module B: working-day calc, SLA state, analytics queries
│   │       ├── evc/              # Module C
│   │       ├── undertakings/     # Module D
│   │       ├── documents/        # Module E: upload, scan, extraction orchestration, verification
│   │       ├── notifications/    # Module F: rules, templates, delivery
│   │       ├── reports/          # Module F: R1–R6, case summary PDF
│   │       ├── audit/            # Hash-chain writer + verifier + anchoring
│   │       ├── auth/             # OIDC, RBAC, scope, step-up
│   │       ├── clock/            # Virtual clock service
│   │       ├── adapters/         # AICTE portal, identity, messaging, eSign, DigiLocker (mock + interface)
│   │       └── demo/             # Time travel, reset, proof panel
│   └── worker/                   # Sweeper, BullMQ processors, OCR/PDF jobs, daily anchor
├── packages/
│   ├── shared/                   # Zod schemas, DTO types, event names, constants
│   ├── workflow/                 # Pure TS: definition schema, validator, XState builder, rules wrapper (unit-testa
│   ├── db/                       # Prisma schema, migrations, scoped client, seed
│   └── ui/                       # SlaBadge, EscalationTimeline, PizzaTracker, RuleDiffModal, CaseSummaryModal …
├── tests/
│   ├── e2e/                      # Playwright: the demo script as an automated test
│   └── simulation/               # 365-day, 5,000-record deadline reliability simulation
└── docs/                         # OpenAPI export, ADRs, runbook, PPT assets, demo video script
```

---

11. Seed Data Specification (deterministic — same data every run)

Use `faker` with a **fixed seed**, plus hand-authored "story" records so the demo is predictable. Targets: **9 regions, ~60 institutions, ~150 applications, ~40 undertakings, ~120 tracked documents**, plus a larger **5,000-record synthetic set** loaded only for the simulation/performance tests.

Scenario (story record)	State at seed time	Used in
Institution A — New Institution, Scrutiny+EVC track	In `SCRUTINY`, SLA at ~60%	Demo step 2 (revert)
Institution B — New Institution, Scrutiny+EVC track	In `SCRUTINY`, clean file	Demo step 2 (clear → auto-jump to EVC queue)
Institution C — EoA, Only Scrutiny	`AMBER` (82% SLA)	HQ dashboard filter
Institution D — Increase in Intake, EVC+SAC/SHC	`SAC_SHC_REVIEW`, `RED` (112%)	Escalation level 1
Institution E — Only EVC	EVC scheduled, report due in 2 days	EVC alerts
Institution F — Scrutiny file	`CRITICAL` (131%)	Escalation level 2 to Bureau
Institution G	`APPROVED_WITH_UNDERTAKING` — faculty-recruitment undertaking, due in 5 days	Demo step 3 (time travel)
Institution H	Barrier-free certificate undertaking, **already 6 days overdue**	Breach path
Institution I	Registered lease expiring in 28 days; renewal upload ready (sample PDF)	Demo step 4
Institution J	Lease **expired 12 days ago**, `COMPLIANCE_DEFICIENT`	Expired-doc escalation
Institution K	Fire-safety certificate renewed late (history shows catch-up alert)	Audit & reporting
Institution L	Deficiency reverted 9 days ago, response window nearly over	Institution response ladder
Edge cases	Reassigned officer; holiday inside an SLA window; paused-SLA application; low-confidence OCR document; duplicate file hash across two institutions	Tests & Q&A

Also generate **sample synthetic PDFs** (lease deed, fire certificate, evidence photos with EXIF) in `docs/sample-docs/` — one clean, one scanned/rotated (OCR), one with a mismatching date (flag demo). All documents are **synthetic** and watermarked "SAMPLE – NOT A REAL DOCUMENT".

Seeded users (password `Demo@123`, TOTP seed pre-shared for demo): `hq.head`, `hq.officer`, `ro.south`, `ro.west`, `scrutiny.south1`, `evc.lead1`, `sacshc.member1`, `inst.a` … `inst.l`, `auditor`, `admin`.

---

12. Single-Phase Execution Plan (One Build, All Workstreams in Parallel)

There are no phases and no phase gates. Six people start at the same moment, work on one trunk, integrate continuously, and finish together against one Definition of Done. Ordering *inside* a workstream follows technical dependency only; ordering *between* workstreams is removed by **contract-first parallelism** (12.3). The idea-submission PPT (Section 14) is produced by the same single build — Workstream 6 drafts it from the first minute and it is submitted as soon as it passes review, which should be well before 28 October.

12.1 Team ownership (6 members, each owns a vertical slice end-to-end)

ID	Owner focus	Owns (backend + UI + tests)
WS1	**Workflow & Rules core**	`packages/workflow`, definition validator, XState builder, routing rules, transition service, revert/resubmit, HITL guards, SLA working-day math (`packages/workflow` + `apps/api/workflow`, `apps/api/sla`), workflow-definition tests
WS2	**Platform, data & audit**	Monorepo scaffold, Docker Compose, Prisma schema + migrations + immutability trigger, scoped client, auth/RBAC/step-up, audit hash-chain writer/verifier/anchor, virtual clock, seed + simulation dataset, CI pipeline
WS3	**Scheduler, notifications & escalation**	Sweeper, `nextAlertAt` maintenance, escalation matrix execution, notification rules/templates (EN/HI), BullMQ delivery + retries + DLQ, WebSocket alerts, digest mode, Time-Travel endpoint, 365-day simulation test
WS4	**Compliance, documents & reports**	EVC scheduler logic, undertaking ledger, tracked-document register, upload/scan/extraction pipeline, officer verification, case summary PDF, reports R1–R6 + XLSX/PDF exports
WS5	**Dashboards & officer experience**	Bureau HQ Command Center, Regional Desk, My Work, application detail/timeline, EVC board & calendar, region map, bottleneck/workload charts, saved views
WS6	**Institution portal, admin, audit UI, proof & presentation**	Institution portal (pizza-tracker, deficiency responses, uploads, expiry calendar), Policy Configurator UI, audit viewer + tamper demo, Proof Panel, Playwright demo-as-test, **PPT, demo video, README, judge Q&A deck**

(With fewer than six people, merge WS5+WS6 UI work and WS3+WS4 backend work; nothing else changes because the workstreams only touch each other through contracts.)

12.2 Workstream task lists (all run concurrently; "needs" = a contract, not another person's finished feature)

WS1 — Workflow & Rules core

1. Definition JSON schema + validator (reachability, dead ends, targets, terminal, statutory roles). *Needs: none.*
2. XState v5 machine builder from a definition; guard factory (role, scope, step-up, statutory flag, system-actor ban). *Needs: validator.*
3. Routing rules wrapper with fired-rule capture and dry-run endpoint. *Needs: none.*
4. Transition service: single transaction = state change + stage history + audit event + outbox + `nextAlertAt` recompute. *Needs: audit service interface (contract from WS2).*
5. Revert/resubmit/clarification flows, deficiency items, SLA pause/resume. *Needs: transition service.*
6. Working-day calculator (holidays, regional holidays) + property-based tests with fast-check. *Needs: none.*
7. Author the 4 required definitions + 1 extra custom example (for the live "add a track" demo). *Needs: validator.*
8. Exhaustive transition-table tests; export the transition table as a Markdown/PNG for the PPT. *Needs: definitions.*

WS2 — Platform, data & audit

1. Monorepo + Docker Compose + Makefile + `.env.example`; `make up` boots the full stack. *Needs: none.*
2. Prisma schema (Section 8) + migrations + audit immutability trigger + DB roles. *Needs: none.*
3. Auth: OIDC (Keycloak realm export), JWT, RBAC decorators, scope injection (scoped Prisma client), TOTP step-up endpoint. *Needs: schema.*
4. Audit service: canonical JSON, per-stream advisory lock, chain write, verify endpoint, Merkle anchor job. *Needs: schema.*
5. `ClockService` + `/demo/*` guards. *Needs: none.*
6. Seed script (story records + faker set) and 5,000-record simulation seeder; `make seed`, `make simulate`. *Needs: schema.*
7. CI (typecheck, lint, unit, integration with Testcontainers, Playwright smoke) and auto-deploy of the live demo. *Needs: scaffold.*
8. Publish OpenAPI from the API; generate **MSW mock handlers** from Zod schemas for the UI workstreams (this is what removes cross-team blocking). *Needs: shared schemas.*

WS3 — Scheduler, notifications & escalation

1. Notification rule/template/delivery pipeline with `dedupeKey`, BullMQ retry, DLQ; Mailpit email, SMS logger, in-app WebSocket. *Needs: schema.*
2. Sweeper (batch, `FOR UPDATE SKIP LOCKED`, catch-up logic, `isCatchUp`), heartbeat. *Needs: schema + clock.*

3. Escalation matrix executor (levels, acknowledgement-required re-notify). *Needs: rules tables.*
4. EN/HI template set for every event in the catalogue. *Needs: none.*
5. Time-Travel endpoint (`advance-time` + immediate sweep) and the UI control contract. *Needs: clock.*
6. 365-day simulation test with outage injection; Proof Panel metrics endpoint (alert lag, missed = 0, duplicates = 0). *Needs: sweeper + simulation seed.*
7. Daily digest + weekly scheduled reports trigger. *Needs: WS4 report endpoints (contract).*

WS4 — Compliance, documents & reports

1. EVC scheduler: auto-queue on-enter action, priority score with breakdown, mode, committee, conflict check, reschedule history, deadline fields. *Needs: workflow `onEnter` contract.*
2. Undertaking ledger: creation from condition library on approval, lifecycle, evidence upload, verify/reject, extension request, breach + show-cause queue item. *Needs: workflow `onEnter` contract.*
3. Document vault: presigned upload, ClamAV scan, SHA-256, versioning, watermarked download. *Needs: schema.*
4. Extraction pipeline: `RegexBaseline` provider first (offline, deterministic), then `TesseractOCR`, then optional `GeminiFlash`; per-field confidence + source span; confirm form API; cross-check flags. *Needs: vault.*
5. Renewal flow and expiry statuses, `COMPLIANCE_DEFICIENT` propagation. *Needs: register.*
6. Case Summary PDF (Puppeteer template, DRAFT watermark, content hash, finalise action). *Needs: application read model.*
7. Reports R1–R6 with XLSX + PDF export. *Needs: aggregates.*
8. Synthetic sample documents (clean, scanned, mismatching date, duplicate hash). *Needs: none.*

WS5 — Dashboards & officer experience

1. Design tokens, `SlaBadge` (colour + icon + text), layout shell, role-aware navigation, EN/HI toggle. *Needs: none.*
2. Bureau HQ Command Center: region map, funnel, bottleneck ranking, workload chart, Red/Critical feed, saved views. *Needs: aggregate API contract (MSW mocks).*
3. Regional Desk: queue, escalations, reassign/delegate, EVC board. *Needs: contract.*
4. My Work: SLA-sorted inbox, action bar with statutory step-up modal, deficiency modal with reason codes and checklist. *Needs: action contract.*
5. Application detail: stage timeline with durations, deficiency thread, documents tab, case summary modal, audit tab. *Needs: contract.*
6. EVC calendar and committee assignment UI with conflict warnings. *Needs: contract.*
7. Live updates via WebSocket (optimistic UI + toasts). *Needs: WS3 event contract.*

WS6 — Institution portal, admin, audit UI, proof & presentation

1. Institution portal: pizza-tracker, deficiency checklist with per-item upload, undertaking cards with countdown, document expiry calendar + `.ics` feed, notification inbox. *Needs: contract.*
2. Lease upload screen with PDF viewer, highlighted clause, confidence chips, officer confirm form. *Needs: WS4 extraction contract.*
3. Policy Configurator: workflow designer, SLA matrix, escalation matrix, template editor, holiday calendar, with **diff + rollback + dry-run impact**. *Needs: contract.*
4. Audit viewer: filters, chain status, **Verify chain**, sandbox **Simulate tampering**. *Needs: audit contract.*
5. **Proof Panel** (judge page): alert on-time %, lag distribution, missed deadlines = 0, duplicate alerts = 0, audit chain verified, manual follow-ups avoided, OCR override rate. *Needs: WS3 metrics contract.*
6. Playwright script = the demo (Section 13) so the demo is regression-tested.
7. PPT (Section 14), README, architecture/ER diagrams, 3–4-minute demo video (backup), judge Q&A cheat-sheet, one-page "APH items still to confirm" note.

12.3 Contract-first parallelism (how nobody waits on anybody)

Authored by WS1 + WS2 **as the first items inside their own task lists** (they are ordinary tasks, not a separate phase) and committed to `packages/shared`:

1. **Zod schemas / DTOs** for every entity and endpoint.
2. **Workflow definition JSON schema** and the four tracks' definitions.
3. **Event catalogue** (names + payloads) used by notifications, WebSocket and webhooks.
4. **Audit service interface** (`record(streamKey, event, actor)`), **Clock interface** (`now()`), **Notification interface** (`schedule(...)`).

5. **MSW mock handlers** generated from the schemas, with the seed's story records as fixtures.

Until a real endpoint exists, UI workstreams run against the mock handlers; switching a screen from mock to real is a one-line config change, verified by the same Playwright test.

12.4 Continuous integration rhythm (not phases — just how work is merged)

- **Trunk-based development:** short-lived branches, merge to `main` at least every two hours; CI must be green to merge (typecheck, lint, unit tests, integration tests).
- **Always-demoable `main`:** every merge auto-deploys to the live demo environment; a smoke Playwright test runs after deploy.
- **Contract changes** require a PR touching `packages/shared` and tagging the affected workstream owners.
- **Daily 15-minute sync** (or every ~8 hours in a continuous sprint): blockers, contract changes, next demo-script step to rehearse.
- **Rehearsal-driven finish:** the demo script (Section 13) is run end-to-end by someone who did *not* build it; every failure becomes a ticket. The build is done when the Definition of Done passes — not when a clock says so.

12.5 Dependency map (what can start immediately vs. what consumes a contract)

Workstream	Starts at minute 0 on	Consumes contract from
WS1	validator, working-day math, routing wrapper	WS2 audit interface (mockable)
WS2	scaffold, schema, clock, audit	none
WS3	templates, delivery pipeline, sweeper skeleton	WS2 schema/clock; WS1 events
WS4	sample docs, vault, extractor baseline	WS2 schema; WS1 `onEnter` hooks
WS5	design system, shell, charts with mocks	WS2 MSW mocks; WS3 event names
WS6	portal screens with mocks, PPT, diagrams	WS2 MSW mocks; WS3/WS4 contracts

12.6 Definition of Done (single checklist for the whole build)

Functional

- [ ] All four required tracks route correctly end-to-end, plus one admin-created track added live without redeploying.
- [ ] Revert/resubmit loop works with reason codes, checklist items, SLA pause/resume and institution notification.
- [ ] Clearing scrutiny auto-queues the file for EVC; EVC schedule, committee, conflict check, reschedule and report intake work.
- [ ] Approval with undertaking creates ledger records; evidence upload → verify/reject → close; breach escalates and creates a show-cause review item (no auto-penalty).
- [ ] Lease/certificate upload → extraction with highlighted source → officer confirmation → deadlines set; renewal clears overdue state.
- [ ] Escalation matrix fires at 75 / 100 / 125 % and acknowledgement-required re-notification works.
- [ ] All six reports export to XLSX and PDF; Case Summary PDF generates as DRAFT and can be finalised by an authorised officer.
- [ ] Five role dashboards show only in-scope data; English/Hindi toggle works for UI and notification templates.

Safeguards & quality

- [ ] "System tries to approve" test fails as designed; statutory actions need step-up + remarks + authorised human.
- [ ] Audit chain verifies `OK`; sandbox tamper simulation is detected at the exact broken `seq`; `UPDATE/DELETE` on audit table is rejected by the DB.
- [ ] RBAC scope test (every role × every endpoint) shows zero cross-scope leakage.
- [ ] **365-day simulation** over 5,000 records: 0 missed alerts, 0 duplicate alerts; with a 24-hour simulated outage, all missed alerts are sent once on restart.
- [ ] Dashboard p95 < 1 s at 50,000 applications; live update < 2 s.
- [ ] Uploads: infected test file quarantined, wrong MIME rejected, presigned URL expires.
- [ ] Accessibility check (axe) passes on the main screens; no information conveyed by colour alone.

Delivery

- [ ] `make up && make seed` yields a working system on a clean machine in under 10 minutes.
- [ ] Live demo URL up; **Reset Demo** button restores the seed in < 30 s.
- [ ] Demo script passed twice end-to-end by someone who did not build it; Playwright version green in CI.
- [ ] PPT reviewed, official numbers traceable to APH or labelled "configurable", then **submitted on the SIH portal**.

- [ ] 3–4 minute backup demo video recorded; README, architecture diagram, ER diagram and OpenAPI export in `docs/`.

---

13. The 6-Minute Judge Demo Script (also the Playwright E2E test)

Setup: two browser windows (Officer / Institution), Time-Travel control visible, Proof Panel in a third tab. Keep the **Reset Demo** button handy. Backup: the 3–4 minute video.

1. **The macro picture (30 s) — Bureau Head, National Command Center.** Region map coloured by SLA health; open "Red & Critical" saved view → 3 bottlenecked applications, 2 expired leases, 1 breached undertaking. *Say:* "One screen replaces the follow-up calls and spreadsheets."
2. **Handbook-as-data (45 s) — Admin, Policy Configurator.** Show the SLA matrix; change one stage's duration, review the **diff** and the **dry-run impact** ("turns 14 files amber today"), save. Then add a new custom track in the workflow designer. *Say:* "When the APH changes, AICTE changes a setting, not code."
3. **Routing, revert and auto-queue (75 s) — Scrutiny Officer.** Open Institution A → **Revert for Deficiency** (reason code + checklist) → SLA badge turns to *Paused*, audit entry appears, institution gets a notification. Switch to the Institution window: checklist visible, upload a clarification → file returns to Re-Scrutiny. Back as officer: **Clear** Institution B (a routine, non-statutory action, so no step-up is needed) → the file **jumps into the Regional EVC Queue** with a visible priority-score breakdown.
4. **Time-Travel escalation (75 s) — Institution G.** Undertaking due in 5 days. Click **Time Travel +6 days**: sweeper runs, status flips to `BREACHED_ESCALATED`, email/in-app alert appears, the file lands in the RO's escalation queue with a **Show-cause review** item *awaiting a human decision*. Open the Proof Panel: alert lag in seconds, duplicates = 0.
5. **Smart lease renewal (60 s) — Institution I.** Upload a renewed (scanned) lease PDF → malware scan → extraction with **highlighted clause**, confidence chips → officer confirms (shows the amber low-confidence field being corrected) → new expiry date set → overdue flag cleared → reminder ladder reset.
6. **The safeguard moment (45 s) — Regional Officer.** Attempt `APPROVE` → **step-up TOTP + remarks required**; show the "system tried to approve" test result from CI. *Say:* "Automation can route, remind and draft. Only an authorised human can decide."
7. **Case Summary + audit (30 s).** Generate the **Electronic Case Summary** (DRAFT badge) → finalise → open Audit Viewer → **Verify chain** ✓ → toggle **Simulate tampering** → verification fails at the exact entry.
8. **Close (20 s) — Proof Panel.** Show the evaluation-criteria scorecard: workflow accuracy tests green, alert timeliness, **0 missed deadlines in the 365-day simulation**, audit completeness.

13.1 Likely judge questions — prepared answers

Question	Answer
How do you handle APH changes every year?	Workflows, SLAs, thresholds, reminder offsets, document types and templates are versioned configuration with effective dates, diff, rollback and dry-run — no redeploy.
What if your server is down when a deadline passes?	The sweeper derives due work from the database on every run and on boot; missed tiers are sent once with a catch-up flag. We simulate a 24-hour outage in the test suite.
Can the system approve or reject an application?	No. Statutory actions are blocked for system actors and require an authorised human with step-up authentication and remarks; this is enforced in the state machine and tested.
Why trust the AI/OCR dates?	We don't — extraction is advisory with per-field confidence and the source clause highlighted; the officer's confirmed value drives deadlines, and override rate is tracked. A deterministic offline extractor is the default.
Is the audit trail really tamper-proof?	It is **tamper-evident**: per-stream SHA-256 chains, DB-level append-only enforcement and a daily Merkle anchor mean any edit or truncation is detected.
How will it integrate with AICTE's existing portal?	Adapter interfaces with a documented REST/webhook contract, OIDC SSO, and a bulk Excel import so a Regional Office can onboard its existing data immediately.
How does it scale?	Stateless API/worker containers, indexed `nextAlertAt` queries (O(due items)), Redis queues, Postgres read replicas if needed; tested on 50,000 applications.
Data protection?	RBAC + scope enforcement, encrypted vault, short-lived signed URLs, watermarked downloads, India-resident hosting path, DPDP-aligned retention; demo uses synthetic data.
What about languages and accessibility?	English + Hindi UI and notifications, WCAG 2.1 AA / GIGW-aligned design, never colour-only status.
Where did your official numbers come from?	Every official number is configuration mapped from the APH; the placeholders in the prototype are labelled and replaceable via the configurator.

14. Ready-to-Use SIH Idea Submission PPT Content

> Open the **official SIH 2026 template** first and map the content below to its headings (the usual headings are: *Proposed Solution*, *Technical Approach*, *Feasibility & Viability*, *Impact & Benefits*, *Research & References*). Where the template allows fewer slides, merge Slides 2+3 and Slides 6+7. Keep text short on the slides themselves; the detail lives in this plan and the demo.

Slide 1 — Title & Proposed Solution

- **Problem Statement ID:** SIH26262 | **Title:** Integrated AICTE Approval Workflow Automation, Compliance Monitoring and Alert Management System
- **Solution: PRAVAH-AICTE** — *Platform for Regulatory Approval, Vigilance, Automation & Handbook Compliance*
- **One-line concept:** One configurable platform that routes approvals, watches every deadline, and escalates delays — while leaving every statutory decision with the competent authority.
- **How it addresses the problem:** replaces manual follow-up across Regional Offices and Bureaus with automated routing, SLA clocks, EVC tracking, undertaking and lease/document monitoring, tiered escalation and a complete audit trail.
- **What makes it different:**
  - **Handbook-as-data:** APH tracks, SLA days, thresholds and reminders are versioned configuration, not code.
  - **Deadlines that can't be missed:** deterministic, idempotent sweeper with catch-up after downtime.
  - **Safeguard enforced in code:** system actors cannot take statutory decisions; humans need step-up authentication.

Slide 2 — Technical Approach: Modules A–F

- **A. APH Workflow Engine** — versioned JSON definitions → XState machines; rule-based track assignment (Only Scrutiny, Scrutiny+EVC, EVC+SAC/SHC, Only EVC, others); authorised revert with reasons.
- **B. SLA & Pendency Command Center** — working-day SLA with pause/resume; Green <75 % / Amber 75–99 % / Red 100–124 % / Critical ≥125 %; role-based dashboards; bottleneck and workload analytics.
- **C. EVC Queue & Scheduler** — auto-queue after scrutiny; Physical/Online/Hybrid; committee and conflict check; due/overdue escalation.
- **D. Undertaking Ledger** — conditional approvals become tracked records; evidence upload; officer verification; breach escalation to a human show-cause review.
- **E. Document Validity Monitor** — lease/certificate register; AI-assisted extraction with highlighted source, **human-confirmed**; 90/60/30/15/7-day renewal alerts (configurable).
- **F. Notifications, Escalation, Reports & Audit** — event-driven alerts (in-app, email, SMS; English/Hindi), acknowledgement-based escalation, six reports, hash-chained audit trail.

Slide 3 — Process Flow & Architecture

(Use the Section 4.1 diagram on the left and this flow on the right.)

1. **Intake →** rules engine assigns track + SLA clock starts.
2. **Stage routing →** officer acts; auto-advance; deficiency loop pauses SLA and notifies the institution.
3. **EVC →** auto-queue, schedule, track report; escalate delays.
4. **Decision (human) →** competent authority approves / approves with undertaking / rejects with step-up authentication.
5. **Compliance →** undertaking ledger and document watchers created automatically.
6. **Vigilance →** sweeper sends reminders and escalates overdue items; every event is written to the audit chain.

Slide 4 — Feasibility, Viability & Risks

- **Stack (all proven open-source):** Next.js, NestJS, PostgreSQL, Redis/BullMQ, MinIO/S3, XState, Docker.
- **Working prototype:** all modules, seeded with realistic synthetic data, one-command deployment.
- **Integration path:** adapter layer + OpenAPI + webhooks + bulk Excel import; OIDC SSO.
- **Security & compliance path:** RBAC with scope, encrypted vault, malware scan, DPDP-aligned retention, MeitY-empanelled/NIC hosting with India data residency, third-party security audit before go-live.
- **Key risks → mitigations:**

Risk	Mitigation
APH rules change	Handbook-as-data configurator with versioning and dry-run
Alert fatigue	Digest mode, de-duplication, acknowledgement-based escalation
OCR/AI errors	Advisory only; confidence + source highlight; human confirmation
Integration unavailability	Adapters, bulk import, standalone operation
User adoption	Excel onboarding, bilingual UI, role-focused screens

Slide 5 — Novelty, USP & Regulatory Safeguards

- **No-code APH Policy Configurator** with diff, rollback and impact dry-run.
- **Crash-proof deadline engine:** exactly-once alerts and catch-up after downtime, proven by a 365-day simulation.
- **Tamper-evident audit chain** (per-stream SHA-256 + daily Merkle anchor + append-only database) for SAC/SHC appeals, RTI and court queries.
- **Human-in-the-Loop enforced:** statutory actions blocked for the system and require authorised human + step-up; every system-generated artefact is labelled for verification.
- **Explainability:** every routing decision, priority score and extracted date shows *why*.

Slide 6 — Impact & Expected Outcomes (measurable, not guessed)

- **Less manual follow-up:** every reminder/escalation that is auto-sent replaces a manual call or email — tracked by a live "follow-ups avoided" counter and baselined in a pilot Region.
- **Fewer missed deadlines and expired documents:** tiered alerts, institution-side calendars and escalations; target **zero missed alerts** (demonstrated by simulation) and measurable reduction in lapsed undertakings/leases.
- **Faster, more transparent processing:** stage-wise pendency, bottleneck ranking and workload balance visible to every authorised level; institutions see live status.
- **Accountability:** acknowledgement-tracked escalations and complete audit trail.
- **Better use of staff time:** automation handles routing, reminders and draft summaries so officers focus on scrutiny and decisions.
- **Evaluation-criteria scorecard:** workflow accuracy · alert timeliness · deadline reliability · pending-case visibility · manual-effort reduction · audit completeness (live Proof Panel).

Slide 7 — Research & References

- AICTE Approval Process Handbook 2024-25 to 2026-27 and subsequent corrigenda (including the Dec-2025 corrigendum on EVC/SAC constitution).
- AICTE 2026-27 EoA/LoA and SAC/SHC report formats (stage and undertaking structure).
- Digital Personal Data Protection Act, 2023; CERT-In directions; GIGW 3.0; WCAG 2.1; OWASP ASVS.
- Technical references: XState (statecharts), json-rules-engine, BullMQ, PostgreSQL row-level security, RFC 7807, Merkle-tree audit anchoring.

15. Risk Register (build + demo)

#	Risk	Likelihood	Impact	Mitigation / fallback
R1	Scope too large for the team	Medium	High	Contract-first parallelism; every item has a fallback (below); vertical slices keep `main` demoable at all times
R2	OCR/LLM extraction flaky on stage	Medium	Medium	Default to deterministic `RegexBaseline` + curated sample PDFs; manual-entry form always available
R3	Live network/Redis/SMTP failure during demo	Medium	High	Docker Compose on a local laptop as hot spare; Mailpit local SMTP; 3–4-minute backup video; **Reset Demo** button
R4	Time-travel logic corrupts demo state	Low	High	Virtual clock offset only (no row edits); reset snapshot; Playwright regression of the script
R5	Placeholder SLA numbers mistaken for official ones	Medium	Medium	Visible "PLACEHOLDER" labels in admin UI; slide wording "configurable per APH"; APH-confirmation note
R6	Audit chain breaks under concurrency	Low	High	Per-stream advisory locks, `UNIQUE(streamKey, seq)`, concurrency test with parallel writers
R7	Cross-region data leakage	Low	High	Scoped Prisma client + RLS + role×endpoint leakage test in CI
R8	Timezone/DST/holiday errors in deadlines	Medium	Medium	Store UTC, compute in `Asia/Kolkata`; property tests around weekends/holidays; no DST in India but tests keep logic portable
R9	Alert fatigue judged as a weakness	Medium	Low	Digest mode, dedupe, acknowledgement-based escalation shown in demo
R10	Idea cap (500) reached before submission	Low–Medium	High	Submit the PPT as soon as reviewed; do not wait for the deadline day
R11	Judges ask about real AICTE data/integration	High	Medium	Adapter diagram, bulk import demo, honest "mocked in prototype, contract documented" answer

16. Acceptance Test Catalogue (what "works" means)

ID	Test	Pass criteria
T01	Track coverage	Each of the 4 required tracks (+1 custom) completes from `SUBMITTED` to a terminal stage; illegal transitions return 409/422 and are audited
T02	Definition validator	Dead-end, unreachable, missing-target, missing-terminal definitions are rejected and cannot be activated
T03	Revert & SLA pause	After revert, officer SLA is paused; after resubmit, deadline = original + paused working time (property-tested)
T04	Working-day math	Weekends, national and regional holidays are excluded; boundary cases (Fri→Mon, holiday on deadline) correct
T05	Statutory guard	System actor, wrong role, wrong region, missing step-up, or missing remarks → rejected; correct human → accepted with `ActionSignature`
T06	Sweeper exactly-once	Run sweeper N times in parallel → exactly one notification per `dedupeKey`
T07	Catch-up	Stop worker for 24 simulated hours across several tiers → on restart each missed tier is sent once, `isCatchUp=true`, in chronological order
T08	365-day simulation	5,000 records, clock advanced day by day: 0 missed, 0 duplicate alerts; dispatch lag ≤ one sweep interval
T09	Escalation chain	75 % → Level 0, 100 % → Level 1, 125 % → Level 2; unacknowledged escalation re-notifies; acknowledgement stops the loop
T10	EVC flow	Scrutiny clear → queue entry with priority breakdown; conflict-of-interest member blocked; reschedule requires reason; report overdue escalates
T11	Undertaking flow	Create → alerts at configured offsets → evidence → verify/reject; overdue → `BREACHED_ESCALATED` and show-cause *queue item* (not an issued penalty)
T12	Document flow	Extraction returns fields + confidence + source span; officer override recorded; renewal resets ladder; expiry sets `COMPLIANCE_DEFICIENT`
T13	Upload security	EICAR test file quarantined; disallowed MIME rejected; oversized file rejected; presigned URL expires
T14	Audit chain	Verify returns OK; direct SQL `UPDATE/DELETE` is rejected; sandbox tamper detected at exact `seq`; truncation detected by anchor mismatch
T15	Audit coverage	A test enumerates every state-changing endpoint and asserts an audit event is written
T16	RBAC scope	Every role × every endpoint: no out-of-scope record is readable or writable
T17	Reports	R1–R6 return correct aggregates against a known fixture; XLSX/PDF open successfully
T18	Performance	Dashboard aggregates p95 < 1 s at 50,000 applications
T19	Accessibility & i18n	axe passes on main screens; Hindi templates render with no missing keys
T20	Demo script	Playwright version of Section 13 passes against the seeded stack

Appendix A — Review Findings and Changes from v1

#	v1 issue	Severity	v2 change
1	Assumed **4 Regional Offices** and a 4-value `Region` enum; AICTE operates more (Western, Southern, South Western, South Central, Central, Eastern, North Western, Northern, North Eastern)	High (factual)	`Region` table, seeded with all offices; map built from data
2	Modelled **SHC_Hearing → SAC_Appeal** as two sequential stages; AICTE reports use a combined **SAC/SHC** committee; SAC = Standing Appellate Committee	High (factual)	Single `SAC_SHC_REVIEW` stage with outcomes incl. *recommended with undertaking*; labels configurable; SHC expansion flagged to confirm
3	EVC described as RO-constituted; Dec-2025 corrigendum says **Council-constituted** and **Physical/Online/Hybrid**	Medium	`EvcSchedule.mode`, `constitutedBy` (configurable)
4	**Inconsistent thresholds** (70 % amber in Module B vs 75 % in Module F)	Medium	One 75/100/125 threshold set, stored as `EscalationRule` rows
5	**Enum/track mismatch** (`EVC_PLUS_SAC_SHC` vs `SCRUTINY_EVC_SHC_SAC` ; stages in text missing from `AppStage`) while claiming "zero hardcoding"	High	Stages are string keys from versioned `WorkflowDefinition` ; validator ensures consistency
6	**Tech-stack ambiguity** (NestJS *or* FastAPI; PyMuPDF in a TS file; Next.js API routes vs NestJS)	Medium	Definitive stack and ADR table; all-TypeScript monorepo; OCR in Node with provider interface
7	**EVC auto-queue via DB trigger**	Medium	Service-layer `onEnter` action in the same transaction (testable, versioned with definition)
8	**One delayed BullMQ job per deadline**: lost on Redis flush, duplicated on redeploy, can't follow a virtual clock	High	Deterministic idempotent **sweeper** + `nextAlertAt` + `dedupeKey` ; BullMQ only for delivery
9	**Time-Travel** described as a button without a design	High	`ClockService` virtual clock, `/demo/advance-time` , simulation test
10	SLA ignored **working days/holidays** and **pause** mechanics (schema lacked fields)	High	Working-day calculator, holiday calendar, `slaPausedAt/TotalSec` , property tests
11	**Global hash chain** (race conditions) and "tamper-proof" overclaim	Medium	Per-stream chain, advisory locks, canonical hashing, DB immutability, Merkle anchor, verifier; wording = *tamper-evident*
12	HITL claimed but **not enforced**	High	Statutory flag in definitions, system-actor ban, step-up + remarks, `ActionSignature` , tests
13	**Schema gaps:** no config tables for the configurator, no notification table, no deficiency items, no document versions, no stage history, no committee members, `Institution.adminUserId @unique` forced one user per institution	High	Full schema rewrite (Section 8)
14	**Unsubstantiated claims** ("70% reduction", "zero missed", "100% legal traceability")	Medium	Replaced by measurable metrics, simulation proof and a live Proof Panel
15	OCR/LLM treated as **authoritative** ("eliminates fraudulent date entry") and sends documents to an external API with no governance	Medium	Advisory extraction, confidence + source span, officer confirmation, provider interface with offline default, data-governance section
16	**Integration** reduced to one sentence (PS lists it as a key requirement)	Medium	Adapter layer, OpenAPI, webhooks, bulk Excel import, SSO
17	**No security/compliance section** (DPDP Act, CERT-In, GIGW, hosting, malware scan, RLS)	High	Section 7
18	**No accessibility, bilingual, observability, backup, testing strategy**	Medium	Sections 7.2, 7.4, 12, 16
19	**Reports** mentioned in the PS (6 report types) but not specified	Medium	R1–R6 specification and exports

#	v1 issue	Severity	v2 change
20	**No traceability** from PS requirements/evaluation criteria to features and tests	High	Section 3 matrix + scorecard + Proof Panel
21	**Phased/time-boxed sprint by pairs and hour ranges** that only summed to ~36 h and serialised work	Medium	**Single-phase, contract-first parallel build** (Section 12) with one Definition of Done
22	PPT lacked **feasibility/risks and references** slides and wasn't tied to the template or the 500-idea cap	Medium	Slides 4 and 7 added; template-mapping note; submit-early guidance
23	Demo had **no failure fallbacks** and no proof of the claims	Medium	Backup video, local hot spare, reset button, Proof Panel, Q&A prep (Section 13)
24	**Official numbers presented as fact** (SLA days, lease norms) without source	Medium	`PLACEHOLDER` labelling, "APH items to confirm" list (Section 2)